Tenth Grade Tile Task

Standard G.C.2

Name: _____ Date: _____

10

1. a) Inscribed angle that cuts an arc of 18°: angle? ☐

b) Inscribed angle that cuts an arc of 10°: angle? ☐

c) Inscribed angle that cuts an arc of 4°: angle? ☐

2. a) Inscribed angle that cuts an arc of 2°: angle? ☐

b) Inscribed angle that cuts an arc of 8°: angle? ☐

c) Inscribed angle that cuts an arc of 16°: angle? ☐

3. a) Inscribed angle that cuts an arc of 12°: angle? ☐

b) Inscribed angle that cuts an arc of 14°: angle? ☐

c) Central angle 0°: half of it is the inscribed angle? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.C.5

Name: _____ Date: _____

10

1. **a)** Circle circumference 6, arc of 1/1 of the circle: length? ☐
   b) Circle circumference 7, arc of 1/1 of the circle: length? ☐
   c) Circle circumference 4, arc of 1/1 of the circle: length? ☐

2. **a)** Circle circumference 9, arc of 1/1 of the circle: length? ☐
   b) Circle circumference 5, arc of 1/5 of the circle: length? ☐
   c) Circle circumference 4, arc of 1/2 of the circle: length? ☐

3. **a)** Circle circumference 3, arc of 1/1 of the circle: length? ☐
   b) Circle circumference 8, arc of 1/1 of the circle: length? ☐
   c) Circle circumference 5, arc of 1/1 of the circle: length? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.1

Name: _____ Date: _____

10

1. a) Sides on a pentagon? ☐

b) Sides on a nonagon? ☐

c) Sides on a square? ☐

2. a) Sides of a ray? ☐

b) Sides on a triangle? ☐

c) Points that define a line? ☐

3. a) Sides of a line? ☐

b) Sides on a hexagon? ☐

c) Sides on an octagon? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.6

Name: _____ Date: _____

10

1. a) Rotational symmetries of a regular hexagon? ☐

b) Rotational symmetries of a regular pentagon? ☐

c) Rotational symmetries of an equilateral triangle? ☐

2. a) Rotational symmetries of a regular octagon? ☐

b) Rotational symmetries of a scalene triangle? ☐

c) Rotational symmetries of a rectangle? ☐

3. a) Rotational symmetries of a square? ☐

b) Rotational symmetries of a regular nonagon? ☐

c) Rotational symmetries of a regular heptagon? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.7

Name: _____ Date: _____

10

1. a) Lines of symmetry of a scalene triangle? ☐

b) Sides on a nonagon? ☐

c) Sides on a quadrilateral? ☐

2. a) Angles that must match for AA similarity? ☐

b) Sides that must match for SSS? ☐

c) Sides on a hexagon? ☐

3. a) Right angles in a right triangle? ☐

b) Sides on an octagon? ☐

c) Congruence criteria in the list SSS, SAS, ASA, AAS, HL? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.8

Name: _____ Date: _____

10

1. **a)** Missing angle of a triangle with angles 91° and 86° ☐
   b) Missing angle of a triangle with angles 91° and 82° ☐
   c) Missing angle of a triangle with angles 92° and 86° ☐

2. **a)** Missing angle of a triangle with angles 86° and 85° ☐
   b) Missing angle of a triangle with angles 81° and 93° ☐
   c) Missing angle of a triangle with angles 91° and 88° ☐

3. **a)** Missing angle of a triangle with angles 82° and 94° ☐
   b) Missing angle of a triangle with angles 94° and 81° ☐
   c) Missing angle of a triangle with angles 81° and 91° ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.CO.9

Name: _____ Date: _____

10

1. **a)** Supplementary angle of 176°

   b) Supplementary angle of 177°

   c) Vertical angle to a 2° angle (degrees)

2. **a)** Vertical angle to a 6° angle (degrees)

   b) Vertical angle to a 8° angle (degrees)

   c) Supplementary angle of 175°

3. **a)** Supplementary angle of 179°

   b) Supplementary angle of 171°

   c) Supplementary angle of 173°

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: TENTH GRADE

Tenth Grade Tile Task

Standard G.CO.10

Name: _____ Date: _____

10

1. a) Triangle angles 86° and 89°: third angle? ☐

b) Triangle angles 88° and 86°: third angle? ☐

c) Triangle angles 85° and 87°: third angle? ☐

2. a) Triangle angles 90° and 81°: third angle? ☐

b) Triangle angles 92° and 85°: third angle? ☐

c) Triangle angles 91° and 82°: third angle? ☐

3. a) Triangle angles 96° and 80°: third angle? ☐

b) Triangle angles 93° and 87°: third angle? ☐

c) Triangle angles 93° and 85°: third angle? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Tenth Grade Tile Task

10

Standard G.CO.11

Name: _____ Date: _____

1. a) Opposite angle of a parallelogram with a 6° angle? ☐

b) Opposite angle of a parallelogram with a 7° angle? ☐

c) Opposite angle of a parallelogram with a 4° angle? ☐

2. a) Opposite angle of a parallelogram with a 2° angle? ☐

b) Opposite angle of a parallelogram with a 8° angle? ☐

c) Opposite angle of a parallelogram with a 1° angle? ☐

3. a) Opposite angle of a parallelogram with a 9° angle? ☐

b) Opposite angle of a parallelogram with a 3° angle? ☐

c) Opposite angle of a parallelogram with a 5° angle? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GMD.1

Name: _____ Date: _____

10

1. **a)** Diameter of a circle with radius 2? ☐

b) Area of a square of side 1? ☐

c) Radius of a circle with diameter 14? ☐

2. **a)** Volume of a cylinder, r=1, height 1 (π≈3)? ☐

b) Volume of a cube with side 2? ☐

c) Circumference with diameter 2 (π≈3)? ☐

3. **a)** Circumference with diameter 3 (π≈3)? ☐

b) Diameter of a circle with radius 1? ☐

c) Radius of a circle with diameter 10? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GMD.3

Name: _____ Date: _____

10

1. **a)** Volume of a prism 3 by 3 by 1 ☐

b) Volume of a prism 1 by 1 by 1 ☐

c) Volume of a prism 5 by 1 by 1 ☐

2. **a)** Volume of a prism 2 by 2 by 2 ☐

b) Volume of a prism 7 by 1 by 1 ☐

c) Volume of a prism 1 by 1 by 4 ☐

3. **a)** Volume of a prism 3 by 1 by 1 ☐

b) Volume of a prism 1 by 1 by 2 ☐

c) Volume of a prism 1 by 6 by 1 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GPE.4

Name: _____ Date: _____

10

1. a) x-coordinate of point (9, 1) ☐

b) x-coordinate of point (5, 2) ☐

c) x-coordinate of point (6, 1) ☐

2. a) x-coordinate of point (2, 0) ☐

b) x-coordinate of point (7, 0) ☐

c) x-coordinate of point (0, 1) ☐

3. a) x-coordinate of point (8, 1) ☐

b) x-coordinate of point (1, 0) ☐

c) x-coordinate of point (4, 0) ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GPE.6

Name: _____ Date: _____

10

1. **a)** Midpoint of (0, 0) and (0, 0): y?

b) Midpoint of (0, 12) and (0, 0): y?

c) Midpoint of (0, 1) and (0, 13): y?

2. **a)** Midpoint of (0, 2) and (0, 4): y?

b) Midpoint of (0, 4) and (0, 0): y?

c) Midpoint of (0, 7) and (0, 3): y?

3. **a)** Midpoint of (0, 7) and (0, 11): y?

b) Midpoint of (0, 0) and (0, 2): y?

c) Midpoint of (0, 16) and (0, 0): y?

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.GPE.7

Name: _____ Date: _____

10

1. **a)** Distance from (0, 0) to (0, 0) ☐

   b) Distance from (0, 0) to (0, 8) ☐

   c) Distance from (0, 0) to (0, 5) ☐

2. **a)** Distance from (0, 0) to (0, 1) ☐

   b) Distance from (0, 0) to (0, 2) ☐

   c) Distance from (0, 0) to (0, 7) ☐

3. **a)** Distance from (0, 0) to (0, 4) ☐

   b) Distance from (0, 0) to (0, 6) ☐

   c) Distance from (0, 0) to (0, 9) ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.MG.1

Name: _____ Date: _____

10

1. a) Sides of a regular 3-gon used as a model? ☐

b) Curved sides on a model of a circle? ☐

c) Straight sides on a model of a circle? ☐

2. a) Sides of a regular 9-gon used as a model? ☐

b) Sides of a regular 4-gon used as a model? ☐

c) Sides of a regular 6-gon used as a model? ☐

3. a) Sides of a regular 7-gon used as a model? ☐

b) Sides of a regular 5-gon used as a model? ☐

c) Parallel sides on a model of a rectangle? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.2

Name: _____ Date: _____

10

1. a) Scale factor from a side of 7 to a side of 14 ☐

b) Scale factor from a side of 2 to a side of 14 ☐

c) Scale factor from a side of 3 to a side of 9 ☐

2. a) Scale factor from a side of 4 to a side of 16 ☐

b) Scale factor from a side of 2 to a side of 18 ☐

c) Scale factor from a side of 3 to a side of 18 ☐

3. a) Scale factor from a side of 1 to a side of 8 ☐

b) Scale factor from a side of 4 to a side of 4 ☐

c) Scale factor from a side of 3 to a side of 15 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.4

Name: _____ Date: _____

10

1. a) Midsegment of a triangle with base 4? ☐

b) Midsegment of a triangle with base 12? ☐

c) Midsegment of a triangle with base 18? ☐

2. a) Hypotenuse of a 3-4-5 triangle? ☐

b) Midsegment of a triangle with base 14? ☐

c) Midsegment of a triangle with base 8? ☐

3. a) Midsegment of a triangle with base 2? ☐

b) Midsegment of a triangle with base 6? ☐

c) Midsegment of a triangle with base 16? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.5

Name: _____ Date: _____

10

1. a) Missing angle of a triangle with angles 84° and 92° ☐

b) Missing angle of a triangle with angles 93° and 85° ☐

c) Missing angle of a triangle with angles 90° and 81° ☐

2. a) Missing angle of a triangle with angles 81° and 99° ☐

b) Missing angle of a triangle with angles 85° and 88° ☐

c) Missing angle of a triangle with angles 86° and 93° ☐

3. a) Missing angle of a triangle with angles 90° and 82° ☐

b) Missing angle of a triangle with angles 95° and 80° ☐

c) Missing angle of a triangle with angles 81° and 96° ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.6

Name: _____ Date: _____

10

1. a) In a 6-8-10 triangle, opposite leg? ☐

b) Tangent of 45°: numerator when legs are 2 and 2? ☐

c) Tangent of 45°: numerator when legs are 1 and 1? ☐

2. a) Tangent of 45°: numerator when legs are 7 and 7? ☐

b) In a 9-12-15 triangle, opposite leg? ☐

c) In a 6-8-10 triangle, adjacent leg? ☐

3. a) In a 3-4-5 triangle, cos numerator (adjacent)? ☐

b) In a 3-4-5 triangle, sin numerator (opposite)? ☐

c) In a 3-4-5 triangle, hypotenuse? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Tenth Grade Tile Task

Standard G.SRT.8

Name: _____ Date: _____

10

1. **a)** Hypotenuse 5, leg 4: other leg? ☐

b) Hypotenuse 10, leg 6: other leg? ☐

c) Area of a square with side 1? ☐

2. **a)** Area of a right triangle with legs 5 and 2? ☐

b) Hypotenuse 10, leg 8: other leg? ☐

c) Area of a right triangle with legs 7 and 2? ☐

3. **a)** Area of a right triangle with legs 2 and 4? ☐

b) Area of a right triangle with legs 2 and 2? ☐

c) Area of a square with side 3? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9